

Brian Jing

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EDUCATION

University of California, San Diego

August 2024 - June 2027

BS, Data Science - GPA: 3.72

Relevant Coursework: Machine Learning Algorithms, Data Structures & Algorithms, Data Management, Data Visualization, Exploratory Data Analysis, Neural Networks, Representation Learning, Signal Processing

EXPERIENCE

AI Project Manager, Data Science Consultant

October 2025 - Present

Eclipse Analytics 

La Jolla, CA

- Fine-tuning **Meta's SAM 2** model for continuous basketball player tracking through frame cuts, partial frame coverage, and substitutions, implementing a stateful memory for statistical tracking of all ~25 players per game
- Led a team of 3 sports analysts to build a **PCA + Ridge Regression** stacking ensemble that reduced 30+ NBA game features to 1 final "clutchness" score for 700+ modern NBA players, while retaining 82% total variance
- Engineered a "game_changer" meta-feature used in PCA that evaluated a player's influence on winning odds in clutch time, by aggregating clutch-time usage rates, scoring impact, and performance under pressure

Founding AI Engineer

August 2024 - Present

POWERCOACH - Real-Time Powerlifting Form Correction App

Irvine, CA / La Jolla, CA

- Engineering a two-branch **PyTorch neural network** pipeline utilizing 5-frame **spatiotemporal data sequences** to simultaneously classify lift phases and form errors for videos of 10 common powerlifting exercises
- Built a **data pipeline** to segment lift phase clips from USA Powerlifting footage, transforming ~10 hours of video into 25,000+ tabular, frame-level data points for fault detection training via **self-trained barbell detection**
- Developed a full-stack real-time application using **Flask** and **WebSockets**, delivering low-latency audiovisual feedback from the AI backend to the **React** frontend through JPEG compression and payload minimization

Data Science Lead

July 2023 - April 2024

Ignited Minds Nonprofit 

Irvine, CA

- Designed and taught the first **data science curriculum** for 11 Malawian high schools remotely, coordinating technical infrastructure funding with local administrators to establish sustainable after-school programs
- Iteratively enhanced teaching methods based on class participation metrics, fostering engaging learning environments that successfully introduced foundational **Python, R**, and data analysis skills to beginners
- Grew a **global community** of 1,000+ aspiring analysts by creating tutorials of data science concepts and technologies in the form of online monthly follow-along blogs

PROJECTS

BFRB Gesture Classifier 

June 2025 - September 2025

- Engineered an **XGBoost + PyTorch** neural network pipeline to classify body-focused repetitive behaviors (BFRBs) based on wrist-worn sensor data, improving testing F1 score by ~40% using **5-fold cross validation**
- Performed multivariate imputation on 30,000+ missing data entries within a 500,000-entry dataset using **scikit-learn**, optimizing imputation hyperparameters with **Bayesian optimization** to preserve feature variance
- Boosted model training efficiency while maintaining classification accuracy by reducing 300 of the dataset's redundant raw sensor variables to 25 **extracted statistical features**

X-Ray Tuberculosis Detector 

June 2024 - July 2024

- Trained a **Keras CNN** on 4,200+ chest X-ray images to diagnose tuberculosis, achieving 98% testing accuracy by iteratively optimizing neural network convergence through **Adam optimization** and **ReduceLROnPlateau**
- Open-sourced the model architecture on Kaggle to drive community innovation, calling on 2,000+ viewers to develop the model further and inspiring deeper exploration into computer vision innovations in healthcare

SKILLS

Deep Learning & Modeling: PyTorch, Tensorflow/Keras, Scikit-learn, XGBoost, Random Forest

Full-Stack Development: JavaScript, Flask, WebSockets, React, AWS (RDS, ECS), SwiftUI

Data Engineering & Visualization: PostgreSQL, R, Data Imputation, OpenCV, Seaborn, Pandas, NumPy